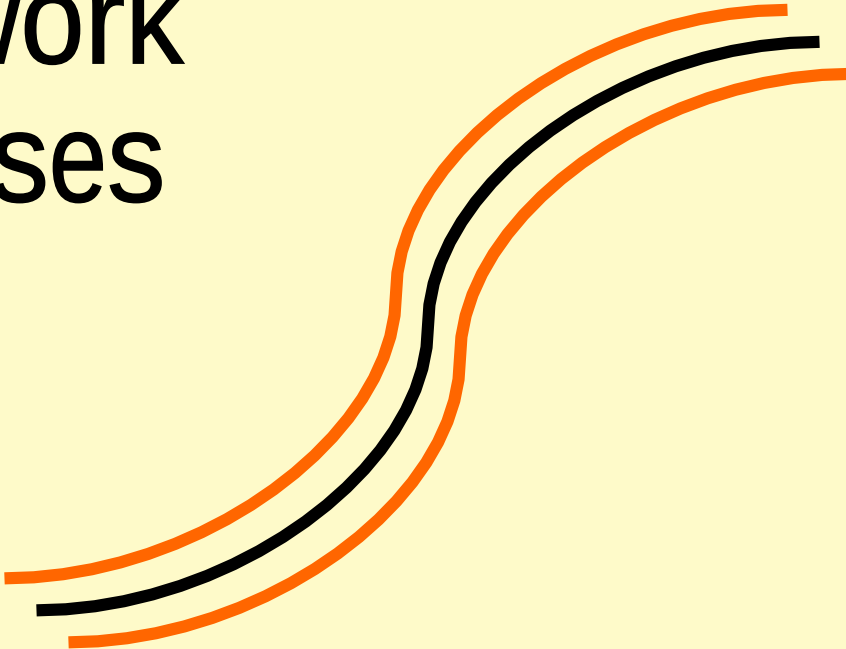




CEG 4424/6424  
Security Attacks & Defenses

Non-TCP/IP Network  
Attacks & Defenses



# Today's Objective(s)

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Understand the basic components of different non-TCP/IP based types of networks along with their most common attacks/vulnerabilities and potential defenses/mitigations

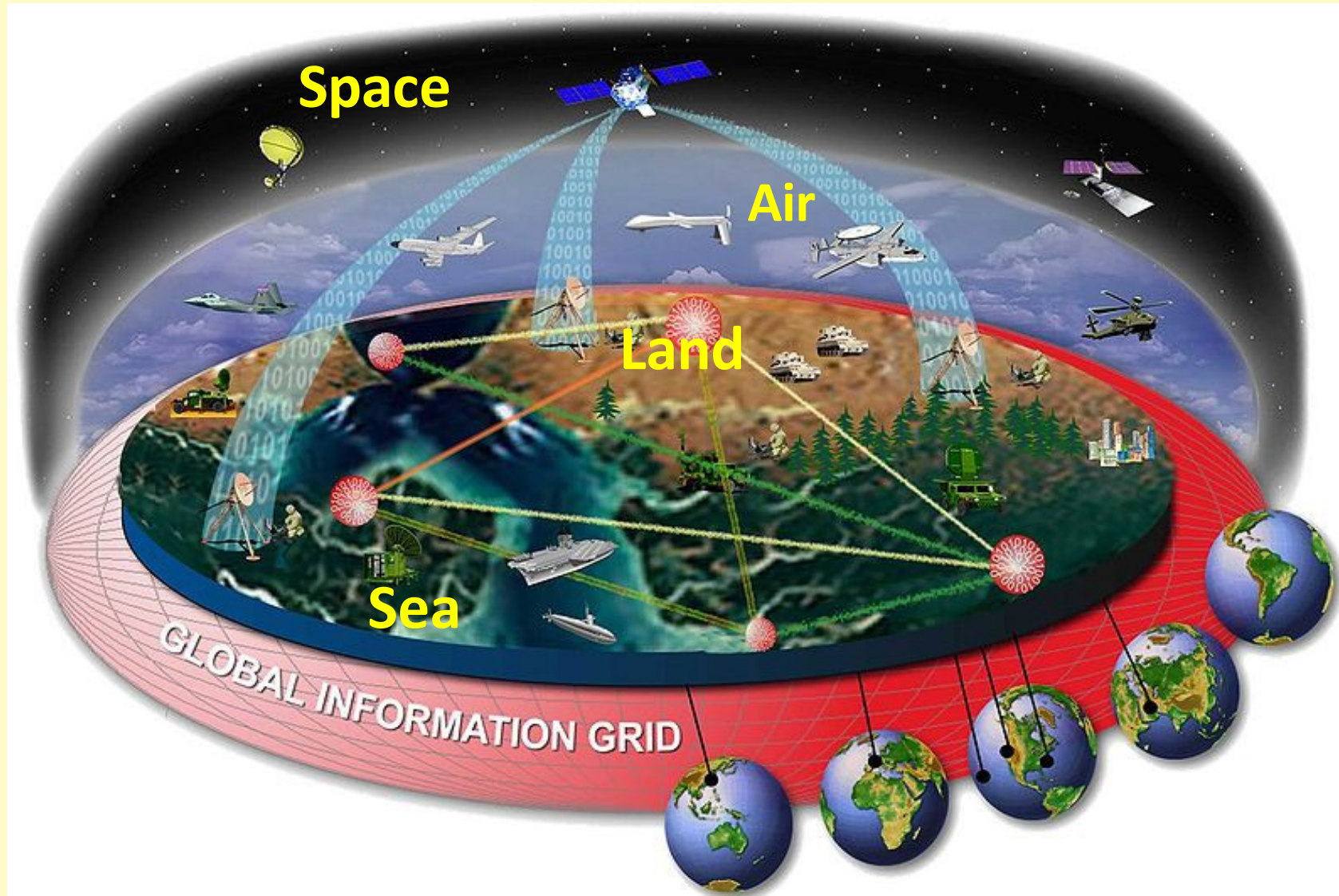
- Mil Standard 1553
- Controller Area Network (CAN)
- Smart Grid

## Disclaimer:

- Slide contents are from numerous sources:
  - Previous CEG 6424 lectures
  - *Experimental Security Analysis of a Modern Automobile*, Karl Koscher, Alexei Czeskis, Franziska Roesner, Shwetak Patel, and Tadayoshi Kohno (Department of Computer Science and Engineering University of Washington) & Stephen Checkoway, Damon McCoy, Brian Kantor, Danny Anderson, Hovav Shacham, and Stefan Savage (Department of Computer Science and Engineering University of California San Diego)
  - *Adventures in Automotive Networks and Control Units* By Dr. Charlie Miller & Chris Valasek
  - Aircraft Hijacking, Practical Aero Series, Hugo Teso, n.runs Professionals -Security Research Team -April 2013
  - Others as identified on individual slides

# CYBERSPACE SYSTEMS OF SYSTEMS (SOS) VIEW

*"Networks, Networked Networks & more Networks"*



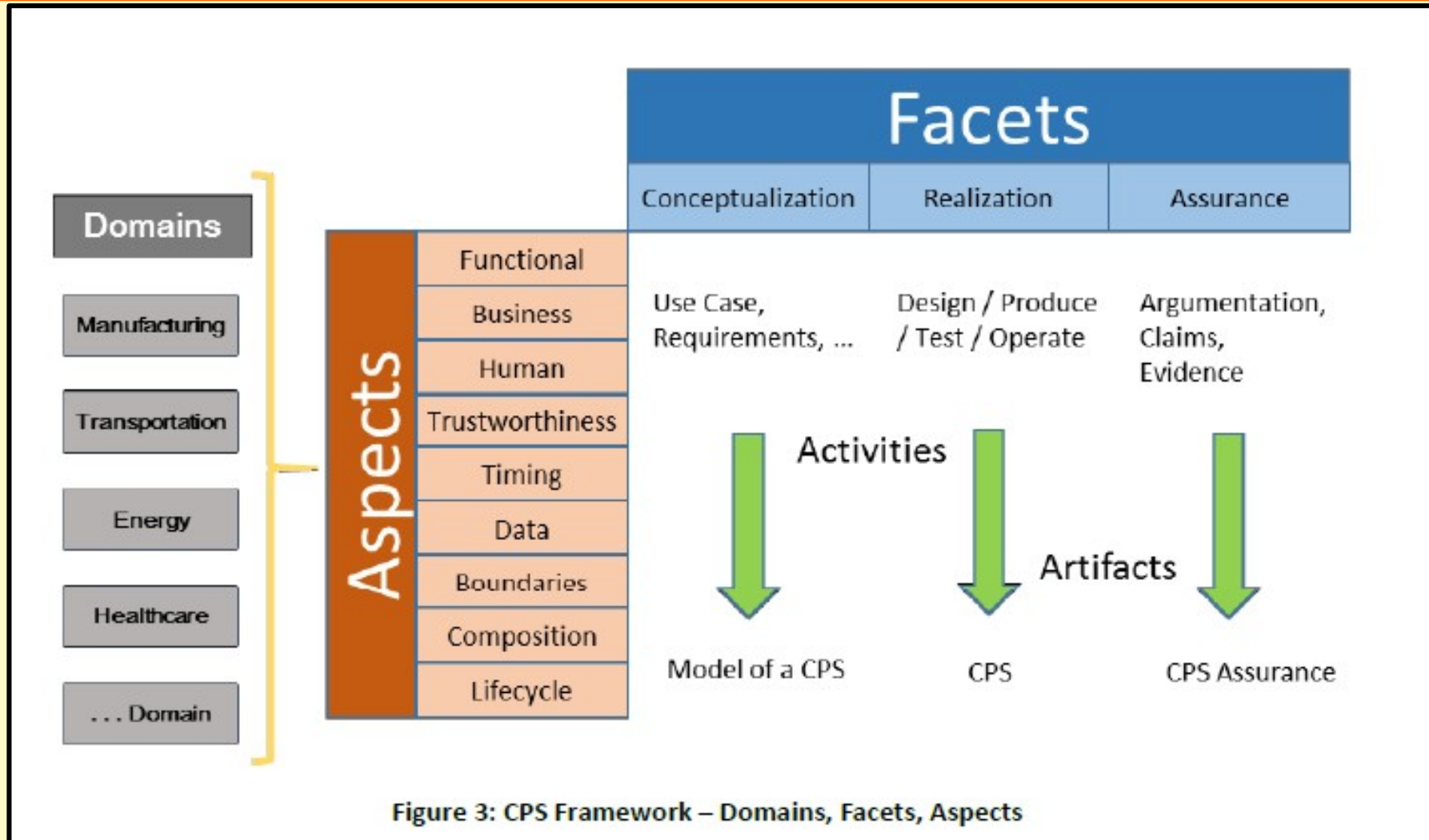
# Cyber-Physical Systems (CPS)

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- Cyber-Physical Systems (CPS) = engineered systems that are built from, and depend upon, the seamless integration of computational algorithms and physical components. Advances in CPS will enable capability, adaptability, scalability, resiliency, safety, security, and usability that will far exceed the simple embedded systems of today. CPS technology will transform the way people interact with engineered systems -- just as the Internet has transformed the way people interact with information. New smart CPS will drive innovation and competition in sectors such as agriculture, energy, transportation, building design and automation, healthcare, and manufacturing.
- We do not yet have a mature science to support systems engineering of high-confidence CPS
- Traditional analysis tools are unable to cope with the full complexity of CPS or adequately predict system behavior
- One barrier to progress is the lack of appropriate science and technology to conceptualize and design for the deep interdependencies among engineered systems and the natural world
- New relationships between the cyber and physical components require new architectural models that redefine form and function
- They integrate the continuous and discrete, compounded by the uncertainty of open environments
- Traditional real-time performance guarantees are insufficient for CPS when systems are large and spatially, temporally, or hierarchically distributed in configurations that may rapidly change
- With the greater autonomy and cooperation possible with CPS, greater assurances of safety, security, scalability, and reliability are demanded, placing a high premium on open interfaces, modularity, interoperability, and verification.

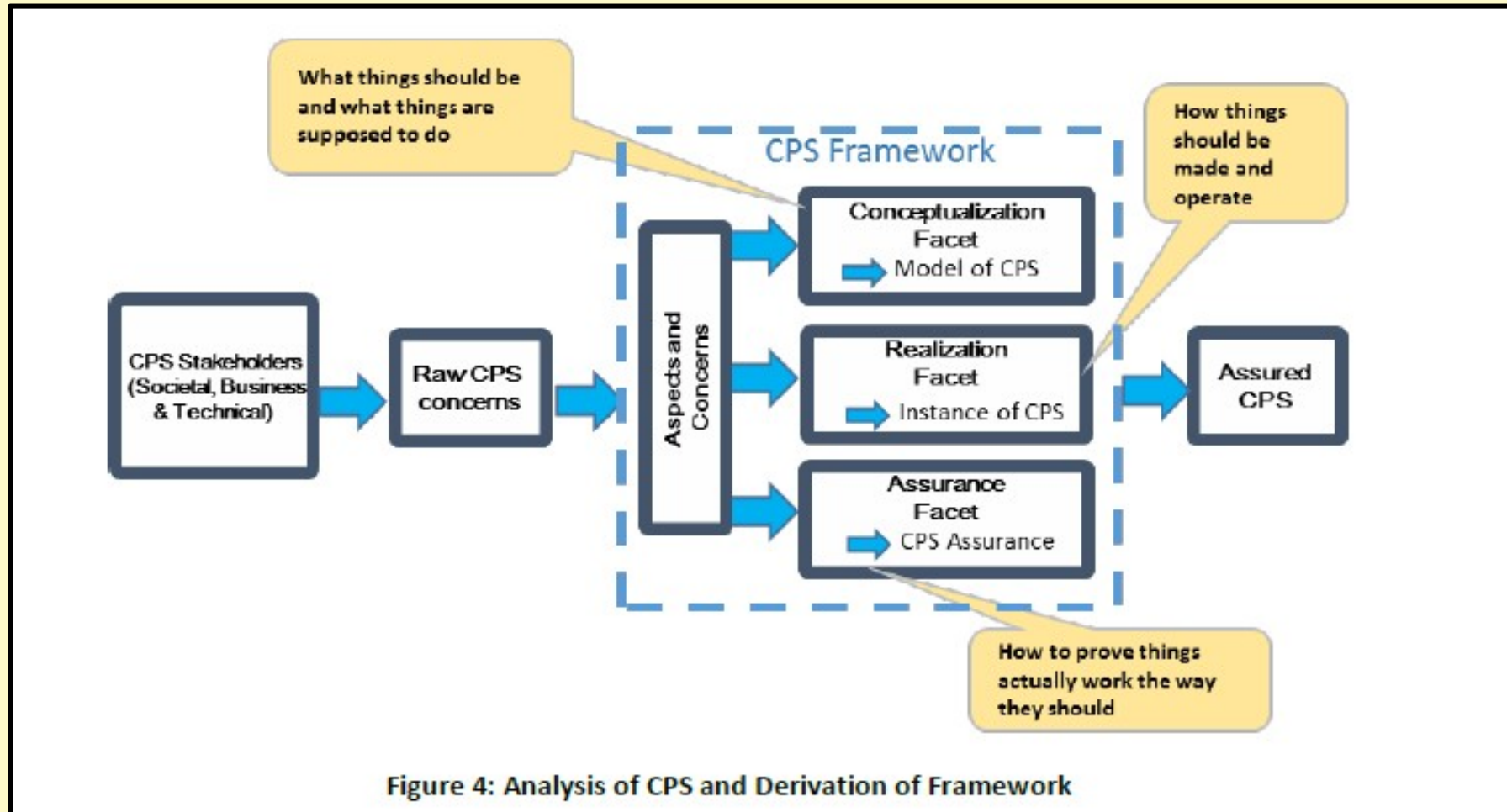
<https://www.nsf.gov/pubs/2016/nsf16549/nsf16549.pdf>

# CPS Framework





# CPS Framework

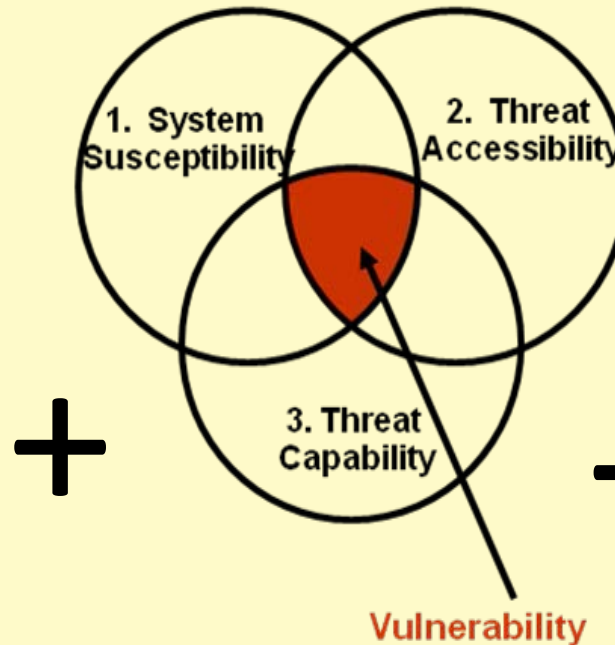
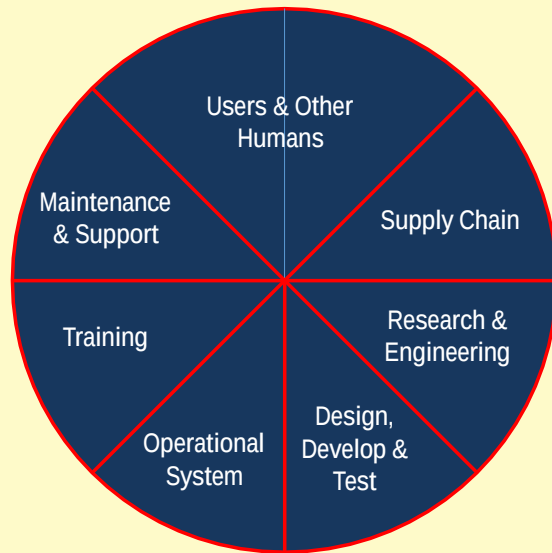




# Unique, Cost Effective Methodology Initially Developed at AFRL



Total system approach, focused on mission assurance, using a unique, validated analysis methodology



3 SPI Tenets - reduce a System's vulnerability by:

- Focus on what is critical – reduce the number of potential susceptibility points
- Move It 'Out of Band' – make the critical pieces less accessible (or inaccessible)
- Detect, React, Adapt – deny the threat the ability to do harm (get inside the adversary's OODA loop)

The Methodology is sound – now need to research & develop the processes, tools & technologies to efficiently implement the methodology



# Required Skill Sets



## 1. System Susceptibility

- System Specific Engineer – expert knowledge of subsystems/component functionality
- Reverse Engineer – expert on determining critical system functionality and possible exploits of that functionality

## 2. Threat Accessibility

- Penetration Tester – expert knowledge of how to “penetrate”/gain access to a system using “Total Weapons System” approach

## 3. Threat Capability

- Intelligence Analyst – expert knowledge of threat capabilities

## 4. Mission Assurance

- Mission Expert – operations expert that knows how the system is used in a mission context



**Multi-Disciplinary Expertise Required for most (if not all) Complex Systems Analysis**



# The Internet and other TCP/IP Based Networks

---

- Required Skill Sets
  - System Susceptibility
    - System Specific Engineer – expert knowledge of subsystems/component functionality – COMPUTER SCIENTIST
    - Reverse Engineer – expert on determining critical system functionality and possible exploits of that functionality – COMPUTER SCIENTIST
  - Threat Accessibility
    - Penetration Tester – expert knowledge of how to “penetrate”/gain access to a system using “Total Weapons System” approach – CYBER SECURITY COMPUTER SCIENTIST
  - Threat Capability
    - Expert knowledge of threat capabilities – COMPUTER SCIENTIST (we built the internet and the web ... we know what capabilities bad guys have)
  - Mission Assurance
    - Mission Expert – operations expert that knows how the system is used in a mission context – DEPENDS ON THE APPLICATION DOMAIN but more often than not – COMPUTER SCIENTIST

# Non-TCP/IP Based Networks

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- TSA Required Skill Sets

## System Susceptibility

- System Specific Engineer – expert knowledge of subsystems/component functionality – NOT COMPUTER SCIENTIST
- Reverse Engineer – expert on determining critical system functionality and possible exploits of that functionality – NOT COMPUTER SCIENTIST

## Threat Accessibility

- Penetration Tester – expert knowledge of how to “penetrate”/gain access to a system using “Total Weapons System” approach – CYBER SECURITY COMPUTER SCIENTIST

## Threat Capability

- Intelligence Analyst – expert knowledge of threat capabilities – TBD - DEPENDS ON THE APPLICATION DOMAIN BUT MOST LIKELY NOT COMPUTER SCIENTIST

## Mission Assurance

- Mission Expert – operations expert that knows how the system is used in a mission context – TBD – DEPENDS ON THE APPLICATION DOMAIN BUT MOST LIKELY NOT COMPUTER SCIENTIST

# What does Cyberspace look like?

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# Mil Std 1553

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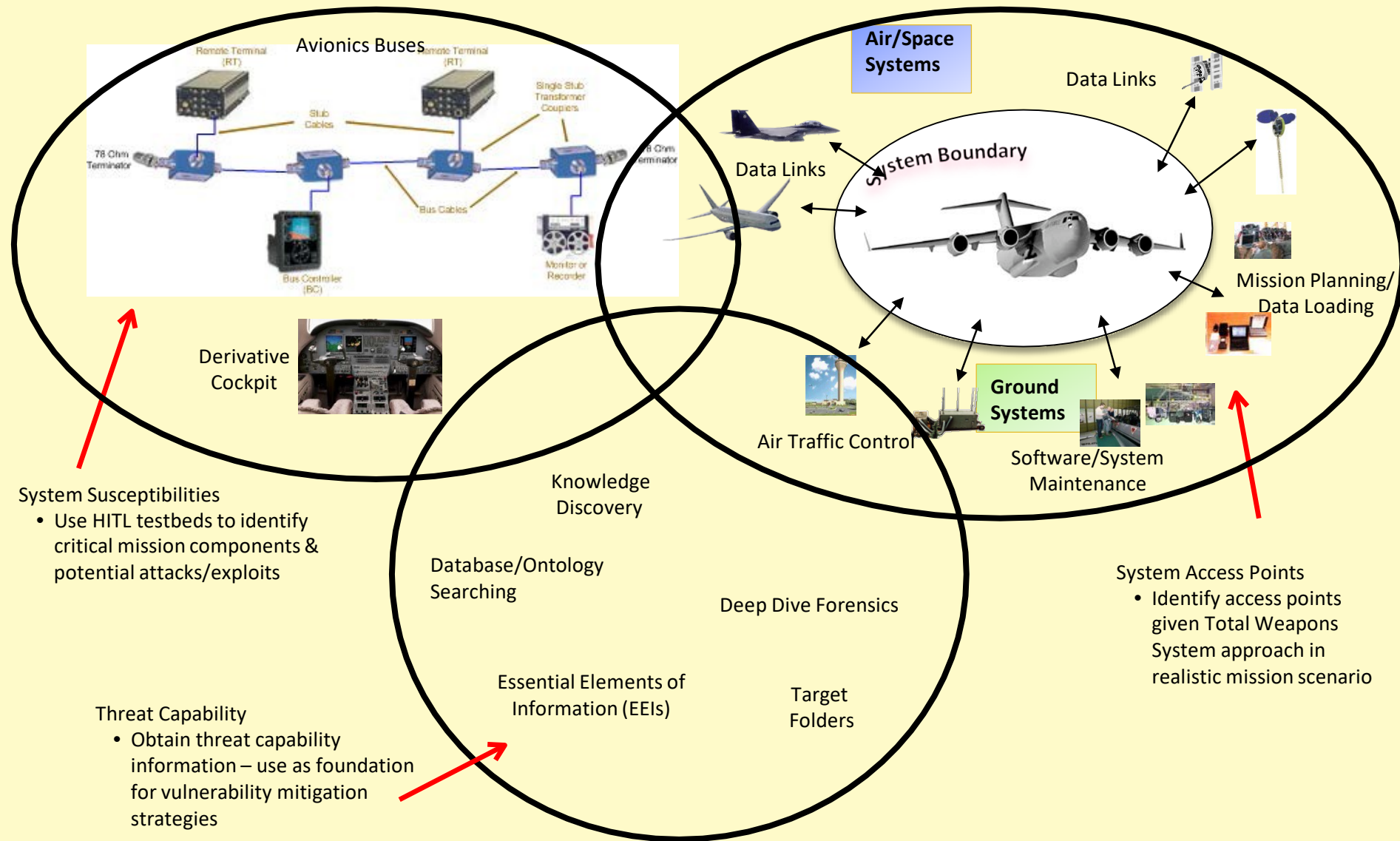
- MIL-STD-1553 is a military [standard](#) published by the [United States Department of Defense](#) that defines the [mechanical](#), [electrical](#), and functional characteristics of a [serial data bus](#).
- It was originally designed for use with military [avionics](#), but has also become commonly used in spacecraft on-board data handling (OBDH) subsystems, both military and civil
- It features a dual redundant [balanced line](#) physical layer, a (differential) [network interface](#), [time division multiplexing](#), half-duplex command/response protocol, and up to 31 remote terminals (devices).
  - A version of MIL-STD-1553 using optical cabling in place of electrical is known as MIL-STD-1773.
- MIL-STD-1553 was first published as a [U.S. Air Force](#) standard in 1973, and first was used on the [F-16 Falcon](#) fighter aircraft
- Other aircraft designs quickly followed, including the [F-18 Hornet](#), [AH-64 Apache](#), [P-3C Orion](#), [F-15 Eagle](#) and [F-20 Tigershark](#).
- It now is widely used by all branches of the [U.S. military](#) and has been adopted by [NATO](#) as [STANAG 3838 AVS](#)
- The Russian made [MiG-35](#) also uses MIL-STD-1553
- MIL-STD-1553 is being replaced on some newer U.S. designs by [IEEE 1394](#) (aka FireWire)

# AIT MIL-STD-1553 *Single-, Dual-, and Quad-Channel MIL-STD-1553 PXI Modules*

- 1, 2, or 4 dual redundant MIL-STD-1553 bus interfaces with full error injection and detection
- Concurrent BC, multiple RT (31), and BM operations for extended function interfaces
- Onboard timetag clock synchronization to external IRIG or PXI system clock
- PXI trigger input/output support (generate PXI triggers on 1553 bus events)
  - PCI extensions for instrumentation - spec defines a rugged PC-based platform for measurement & automatio systems
- Variable output voltage signal and software-selectable bus coupling modes
- Advanced driver for developing applications in LabVIEW, LabVIEW Real-Time, and Windows-based C/C++



# System Vulnerability Model Expanded

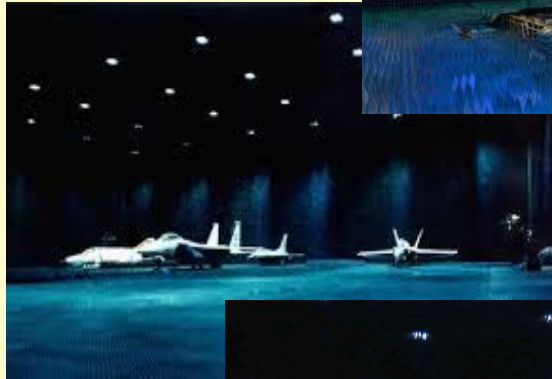




# CPS Analysis & Testing Environments

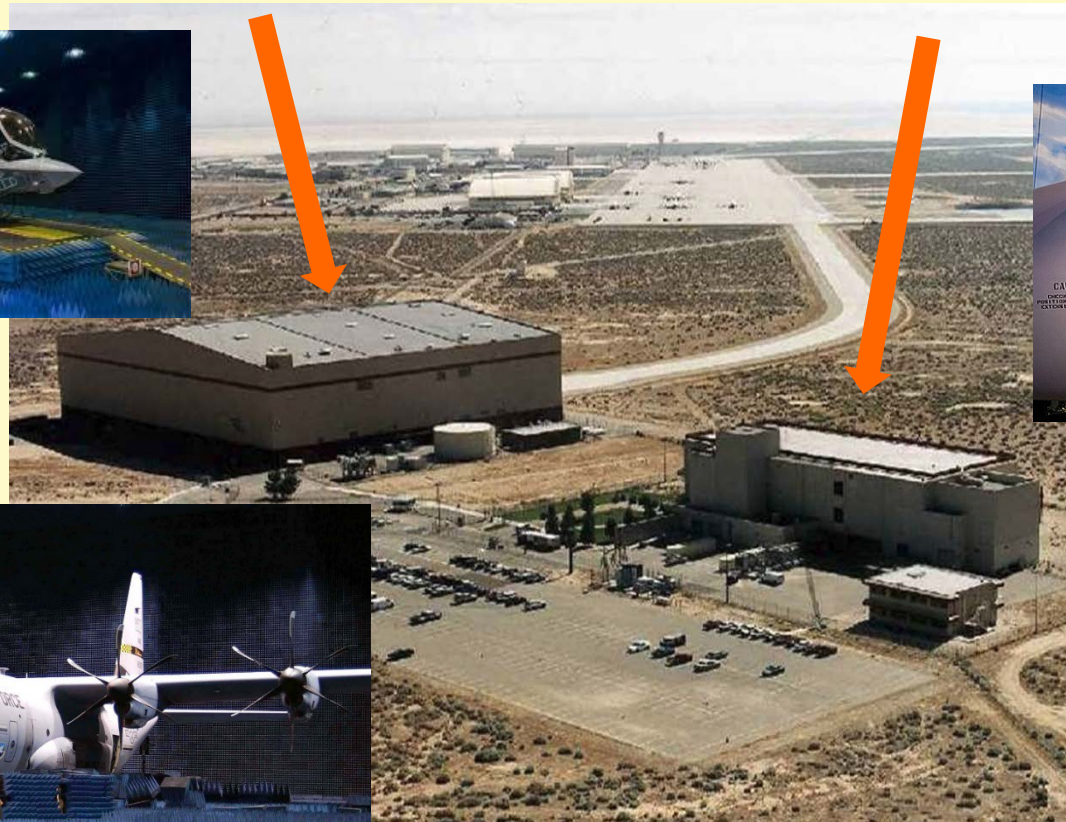


Radiation Absorbent  
Material (RAM)



Benefield Anechoic Facility  
(BAF)

Integrated Facility for Avionics Systems Test  
(IFAST)



# Hacking an Airplane With Only an Android Phone

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- At the [Hack In The Box](#) conference in Amsterdam, security consultant [Hugo Teso demonstrated PlaneSploit](#)
  - An app he developed that he claims can take control of certain systems aboard an airplane and cause it to change direction or just crash itself into the ground.
- Hugo's no terrorist - he developed the app to point out the glaring, frightening, insane security holes in most planes' onboard flight systems
- His demonstration was done in a simulated environment, but the methods and effects, he says, are exactly the same as what could happen with a real plane.
- "The complete attack will be accomplished remotely, without needing physical access to the target aircraft at any time," wrote Teso in his presentation abstract for the conference
- The hack exploits the plane's autopilot, transferring control to the hacker, who (in theory) could command the plane from an app running on Android.

Exercise for students (if interested) = Hugo Teso & Aircraft Hacking

# What does Cyberspace look like?

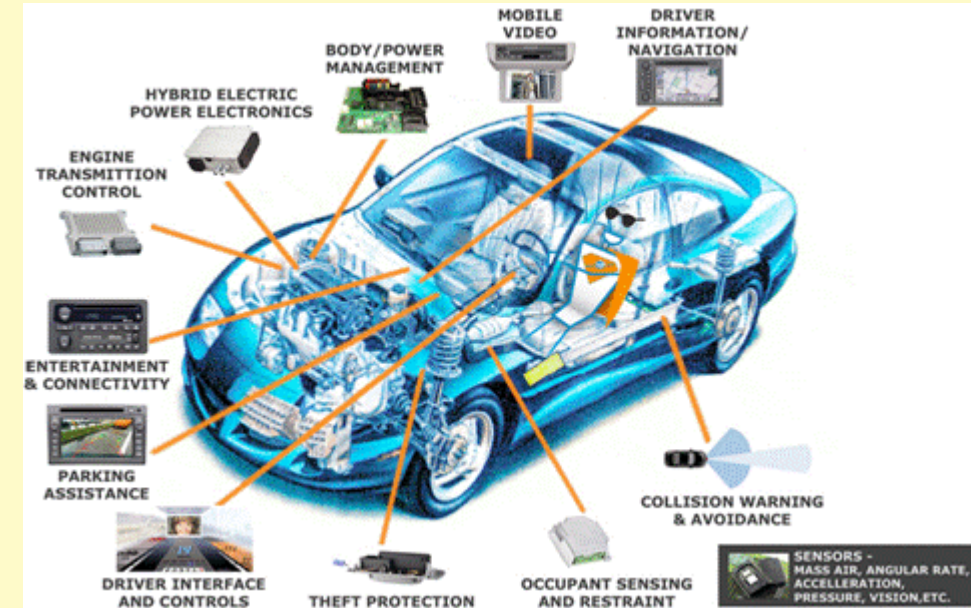
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# Controller Area Network (CAN) Bus

- CAN (Controller Area Network) developed in 1980s for automotive applications
- CAN controller & transceiver chips are reasonably priced due to high quantities required in the automotive industry
  - These facts have made it attractive to developers of non-automotive systems
- Rail vehicles - CAN-based systems are used to connect control devices, sensors or drives
- CAN bus used in space systems of several different countries including the U.S.
- CAN defines physical and data link layers (Layer 1 and 2 of the 7-layer OSI reference model)
  - A higher layer protocol was required (e.g. like TCP was needed for IP)
  - CANopen is a CAN-based higher layer protocol that was developed as a standardized embedded network with highly flexible configuration capabilities
- CAN Protocol:
  - No addresses – uses publish & subscribe communications model
  - CAN ID header indicates packet type
  - All packets broadcast to all nodes
    - Each node determines whether or not to process the packet(s)



# Notes from: *Experimental Security Analysis of a Modern Automobile*

- “the attack surface for modern automobiles is growing swiftly as more sophisticated services and communications features are incorporated into vehicles”
- Conducted a comprehensive analysis of cars’ digital components and internal networks
  - Experimentally evaluate the security properties of each of the key components
  - Analyzed the security properties of the underlying network substrate
- Experimented & analyzed the cars & their internal components ... in three principal settings:

## 1. Bench - physically extracted hardware from the car for analysis

- vehicles use a variant of the Controller Area Network (CAN) protocol for communicating among vehicle components
- both a high-speed and low-speed variant as well as a variety of proprietary higher-layer network management services
- through the CAN protocol, any component can be accessed and interrogated in isolation in the lab

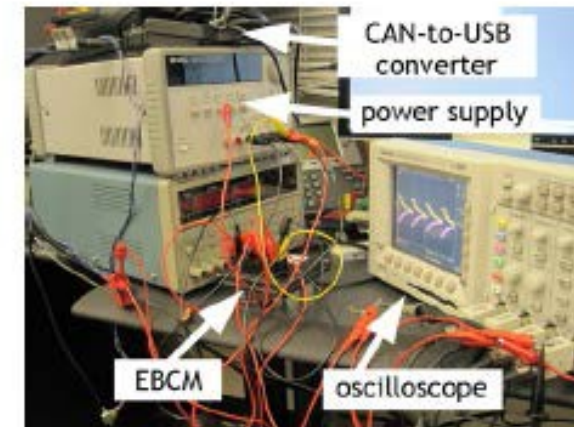


Figure 1. Example bench setup within our lab. The Electronic Brake Control Module (EBCM) is hooked up to a power supply, a CAN-to-USB converter, and an oscilloscope.

# Notes from: *Experimental Security Analysis of a Modern Automobile*

## 2. Stationary car

- Most of in-car experiments conducted with the car stationary
- For both safety and convenience, elevated the car on jack stands for experiments that required the car to be “at speed”; (see Figure 3)
- Figure 2 shows the experimental setup inside the car
  - laptop connected to the standard On-Board Diagnostics II (OBD-II) port
  - Used an off-the-shelf CAN-to-USB interface (the CANCapture ECOM cable) to interact with the car’s high-speed CAN network
  - Used an Atmel AT90CAN128 development board the Olimex AVR-CAN) with custom firmware to interact with the car’s low-speed CAN network.



Figure 3. To test ECU behavior in a controlled environment, we immobilized the car on jack stands while mounting attacks.



Figure 2. Example experimental setup. The laptop is running our custom CARSHARK CAN network analyzer and attack tool. The laptop is connected to the car’s OBD-II port.



# Notes from: *Experimental Security Analysis of a Modern Automobile*

## 3. On the road (full experimental fidelity)

- Some results experimented with at speed while on a closed course.
- Numerous precautions exercised to protect the safety of car's driver and any third parties
  - For example, used the runway of a de-commissioned airport because the runway is long and straight, giving us additional time to respond should an emergency situation arise (see Figure 7)
- For these experiments, one person drove the car while three others drove a chase car on a parallel service road
  - one person drove the chase car
  - one documented much of the process on video
  - one wirelessly controlled the test car via an 802.11 ad hoc connection to a laptop in the test car that in turn accessed its CAN bus.



Figure 7. Road testing on a closed course (a de-commissioned airport runway). The experimented-on car, with our driver wearing a helmet, is in the background; the chase car is in the foreground. Photo courtesy of Mike Haslip.

- Live road tests = “Gold Standard” & were critical for complete understanding of certain functions
- Could not completely characterize the brake behavior until the car was on the road
  - Got illogical behavior when on jack stands because rear wheels were stationary



# Notes from: *Experimental Security Analysis of a Modern Automobile*

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- Some Summary Comments
  - Analysis process important to understand (progressively eliminating “modeling & simulation artifacts or limitations”)
    - Communications protocol analysis
    - Individual component analysis
    - System/sub-system analysis
    - “Operational Environment” analysis
- Past work had indicated potential risks to cyber-physical vehicles – however – never anticipated the ability to create unsafe conditions of such magnitude
- Were surprised at the extent to which the controls that did exist were frequently unused
- CAN Bus security challenges – lots of them
- Deviations from standards – non-compliant access control & imperfect network segregation
- Automobile components clearly & explicitly designed to safely tolerate *failures* – however – tolerating *attacks* has not been part of the same design criteria
  - Believe this is true for all existing CPS & presents 2 challenges
    - Include *tolerating attacks* into the design & development of all future systems
    - Figure out what can be done to existing systems to address the lack of *tolerating attacks* capabilities

# Automobile Hacks & Defenses - Some Examples

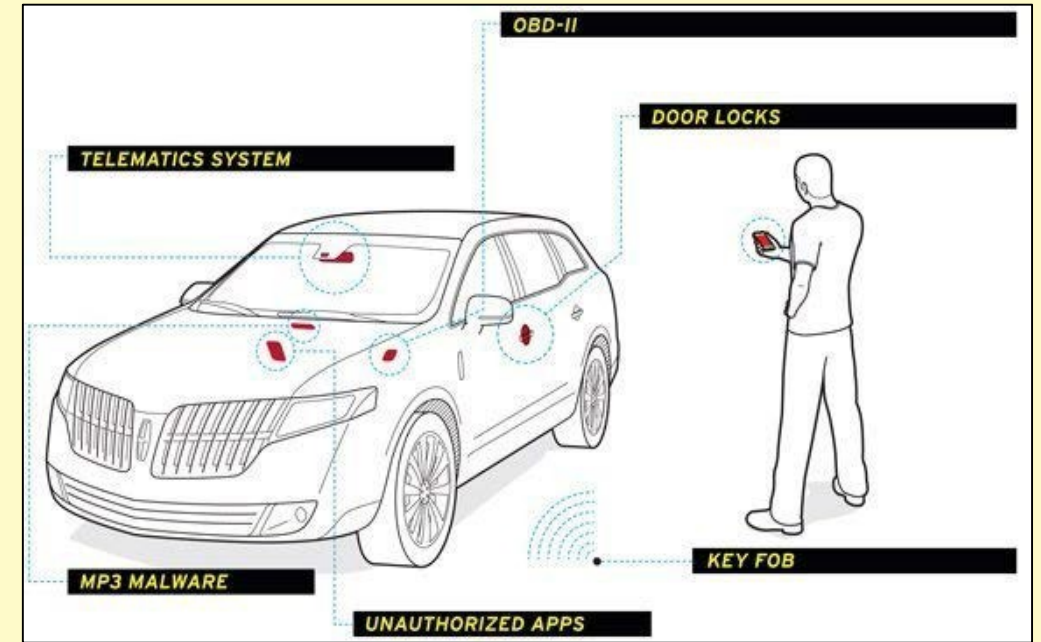
## Vehicle Telematics

- the blending of computers and wireless telecommunications technologies
- provides navigation, communications, safety, security & infotainment.

## TELEMATICS SYSTEM

**THE HACK:** After gaining access to the telematics system, it's possible to control the systems connected to the CAN bus. A hacker could disable a car's ignition the same way an anti-theft system would.

**THE DEFENSE:** Automakers are already beefing up the security of external communications and in-car networks. OnStar, for example, has a "white list" of approved computers that are allowed to connect with cars.



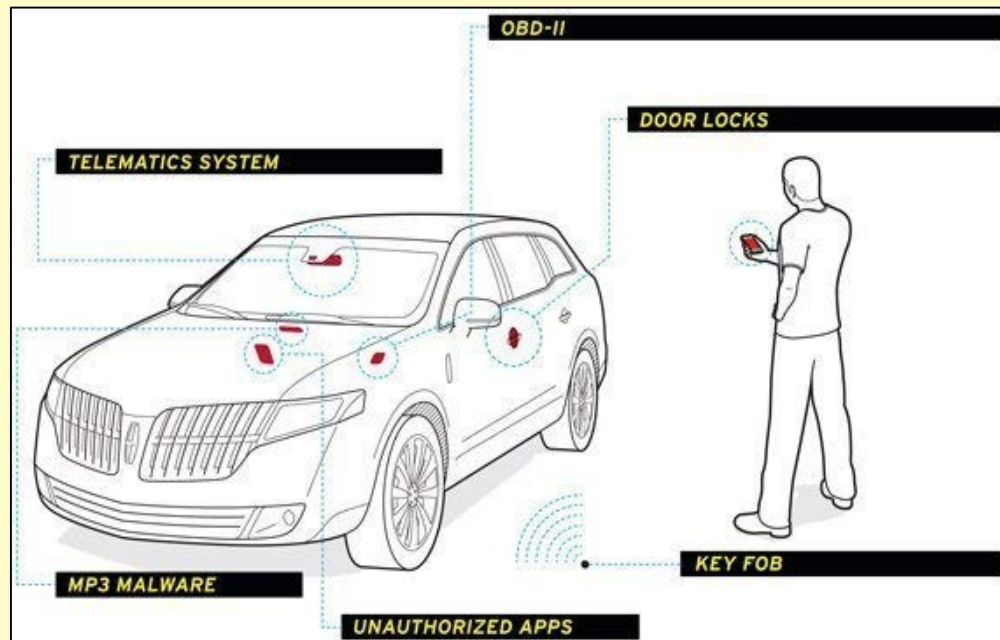
# Automobile Hacks & Defenses - Some Examples

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## MP3 MALWARE

THE HACK: Naughty boy. You downloaded your Odd Future tunes from an unauthorized file-sharing service. Little did you know that version of *Goblin* contains code that battles its way to your car's CAN bus and disables your brakes.

THE DEFENSE: "We harden all our safety-critical systems," says OnStar's security chief Gassenfeit. GM's newer cars, such as the 2011 Chevy Volt, verify any data sent between two systems the same way online retailers process credit cards.



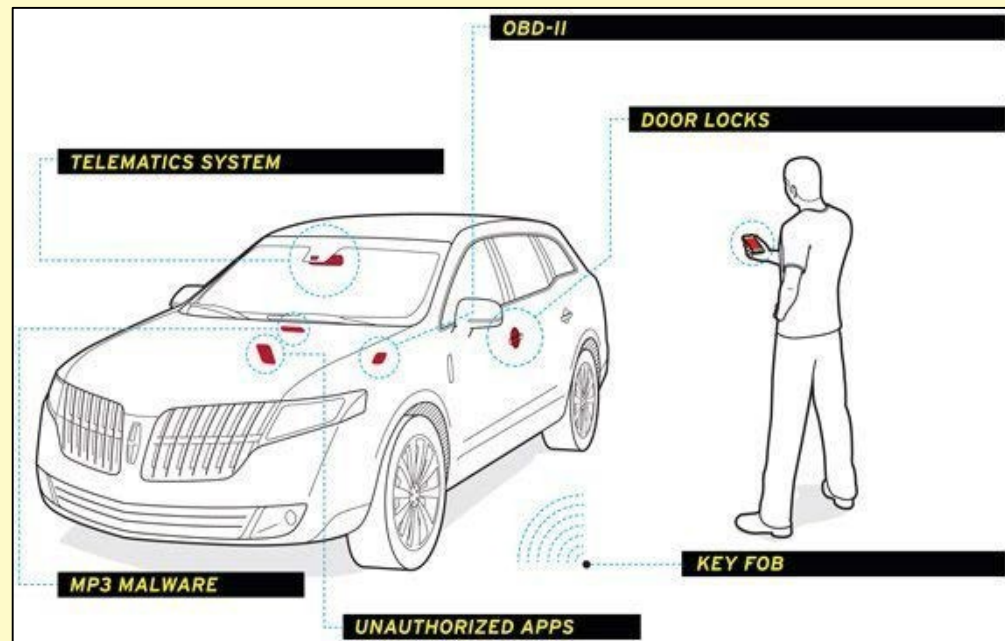
[From the August 2011 Issue of Car and Driver](#)

# Automobile Hacks & Defenses - Some Examples

## UNAUTHORIZED APPS

THE HACK: Carmakers are expanding their infotainment offerings through downloadable software. If a rogue app contains malware or a virus, it can infect your car without your knowledge.

THE DEFENSE: Carmakers are very strict in selecting which apps make it onto their systems. Ford's [MyFord Touch](#) and Toyota's [Entune](#) allow only a handful of preapproved programs, while GM's MyLink goes so far as to route all software through remote servers so that users won't inadvertently install infected apps on their cars.



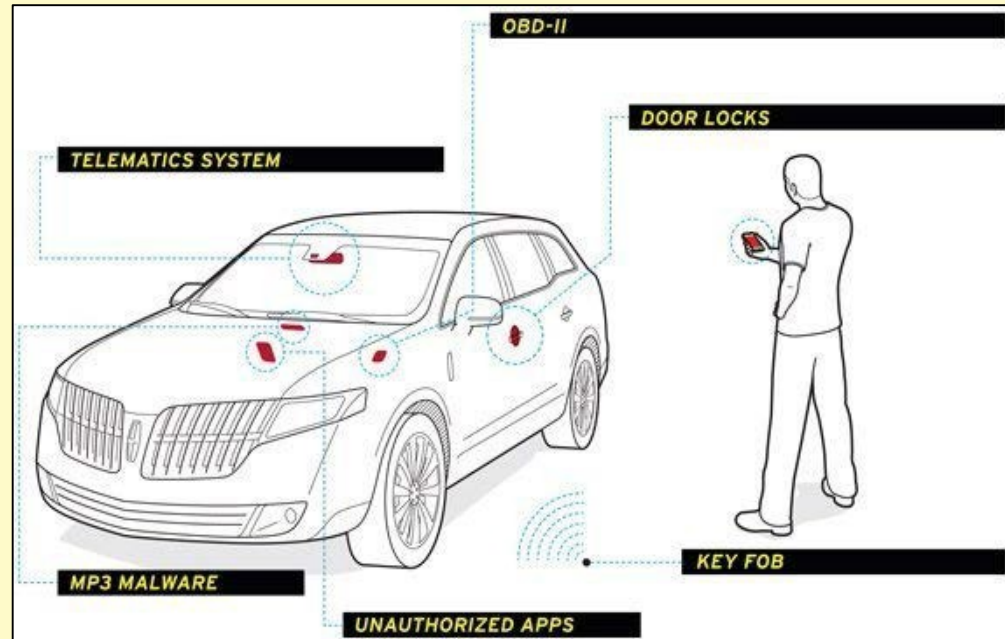
[From the August 2011 Issue of Car and Driver](#)

# Automobile Hacks & Defenses - Some Examples

## *On Board Diagnostics (OBD-II) port*

**THE HACK:** The researchers at CAESS wrote and then installed a program onto the car's CAN bus through the OBD-II port. Once on the network, the program could control every system from the windshield wipers to the brakes. This is the most direct way to hack a car, as it sends code directly to the CAN bus.

**THE DEFENSE:** Carmakers are starting to adopt routine security protocols from the information-technology field, such as protecting files with digital signatures. "What's pretty much standard IT is now being applied to the automotive sector," says Gassenfeit.



[From the August 2011 Issue of Car and Driver](#)



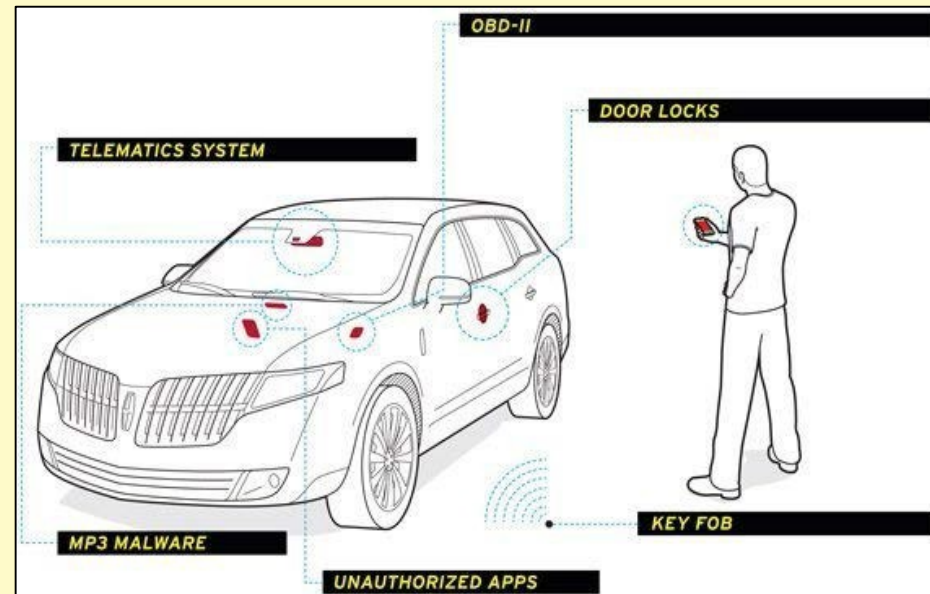
# Automobile Hacks & Defenses - Some Examples

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## DOOR LOCKS

**THE HACK:** In most modern cars, the power-locking mechanism is connected to other vehicle systems so that doors can lock automatically when a car is put into drive and unlock if the airbags have been deployed or the keys are locked inside. If accelerating can engage a car's power locks, a skilled hacker could use the power locks to force that car to accelerate.

**THE DEFENSE:** Infotainment and onboard diagnostic systems are still linked to the module that controls steering and braking. Some carmakers are only allowing the real-time module to send messages in one directions.



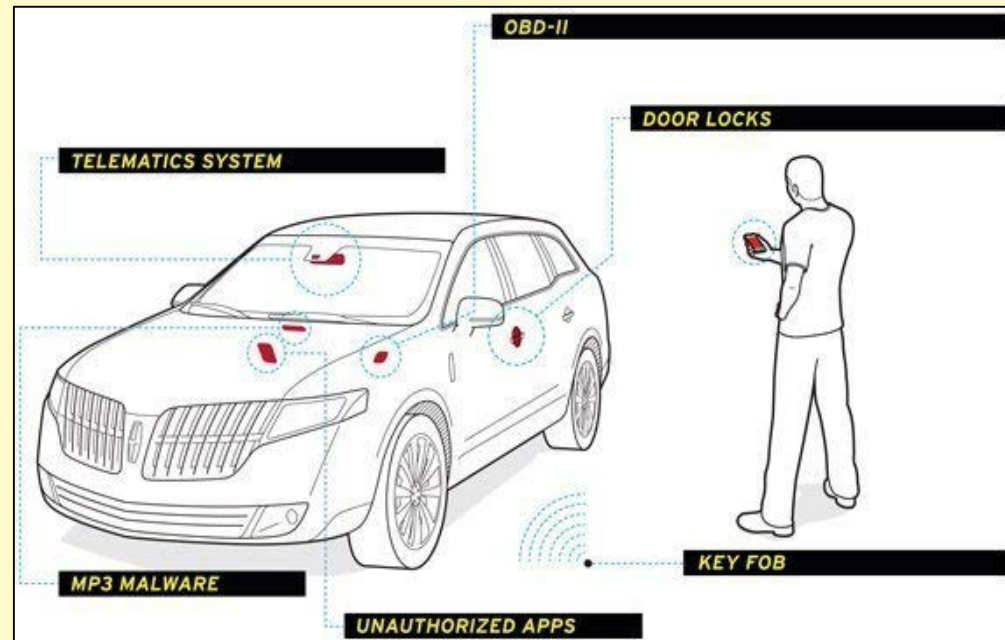
[From the August 2011 Issue of Car and Driver](#)

# Automobile Hacks & Defenses - Some Examples

## KEY FOB

**THE HACK:** Swiss researchers found a way to intercept and extend the key fob signal up to 30 feet with parts that cost less than \$100. The setup doesn't replicate the signal—it just extends its range so the car thinks the key fob is closer than it actually is.

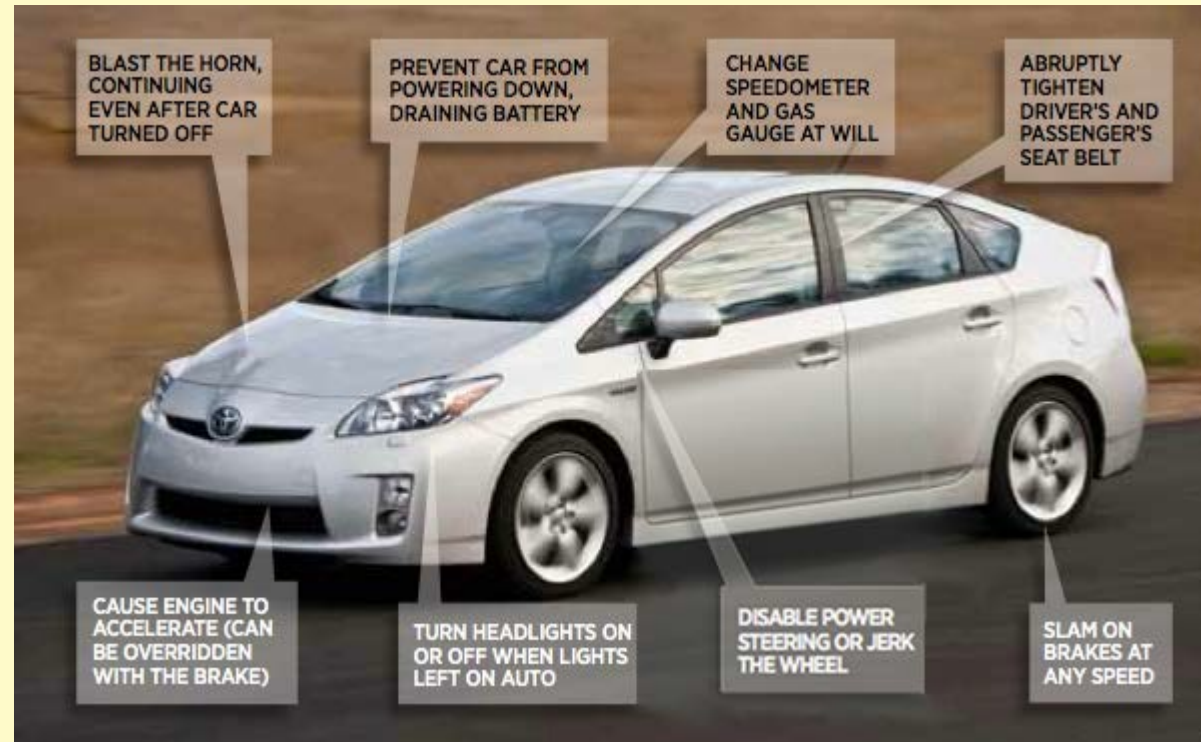
**THE DEFENSE:** There's not much a car manufacturer can do here. These hackers haven't broken the key fobs' encryption in any way—they've just extended its range with a radio repeater. So keep an eye out for anyone loitering in a parking lot and holding a homemade antenna.



[From the August 2011 Issue of Car and Driver](#)

# The Automobile - Hacked

- A car is not a simple machine of glass and steel but a hackable network of computers
  - Charlie Miller, a 40-year-old security engineer at Twitter
  - Chris Valasek, the 31-year-old director of security intelligence at the Seattle consultancy IOActive
    - Received an \$80,000-plus grant last fall from DARPA to root out security vulnerabilities in automobiles



- This story appears in the August 12, 2013 issue of Forbes.

# Notes from: *Adventures in Automotive Networks and Control Units*

## Executive Summary

Previous research has shown that it is possible for an attacker to get remote code execution on the electronic control units (ECU) in automotive vehicles via various interfaces such as the Bluetooth interface and the telematics unit. *(note: they are referring to the work done by the Center for Automotive Embedded Systems Security (CAESS))*

This paper aims to expand on the ideas of what such an attacker could do to influence the behavior of the vehicle after that type of attack. In particular, we demonstrate how on two different vehicles that in some circumstances we are able to control the steering, braking, acceleration and display. We also propose a mechanism to detect these kinds of attacks. In this paper we release all technical information needed to reproduce and understand the issues involved including source code and a description of necessary hardware.

## Introduction

Besides discussing new attacks, this paper aims to bring accessibility to automotive systems to security researchers in an open and transparent way. **Currently, there is no easy way to write custom software to monitor and interact with the ECUs in modern automobiles. The fact that a risk of attack exists but there is not a way for researchers to monitor or interact with the system is distressing.** This paper is intended to provide a framework that will allow the construction of such tools for automotive systems and to demonstrate the use on two modern vehicles. This framework will allow researchers to demonstrate the threat to automotive systems in a concrete way as well as write monitoring and control applications to help alleviate this threat.

# Notes from: *Adventures in Automotive Networks and Control Units*

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- Modern automobiles contain upwards of 50 electronic control units (ECUs) networked together
  - ECUs are essentially embedded devices, networked together on the CAN bus
- There are two main types of CAN packets → normal and diagnostic
  - Normal packets are sent from ECUs and can be seen on the network at any given time
    - They may be broadcast messages sent with information for other ECUs to consume or may be interpreted as commands for other ECUs to act on
    - There are many of these packets being sent at any given time, typically every few milliseconds
  - Diagnostic packets are sent by diagnostic tools used by mechanics to communicate with and interrogate an ECU
    - These packets will typically not be seen during normal operation of the vehicle
- ISO-TP, or ISO 15765-2, is an international standard for sending data packets over a CAN bus, see [\[http://en.wikipedia.org/wiki/ISO\\_15765-2\]](http://en.wikipedia.org/wiki/ISO_15765-2)



# Notes from: *Adventures in Automotive Networks and Control Units*

Ford Escape has 2 CAN Buses

- Medium Speed (MS) operating at 125kbps
  - High Speed (HS) operating at 500kbps
- Both terminate at the OBD-II port = Data Link Connector (DLC) in Ford wiring diagrams

ECUs that reside on the HS CAN bus include

1. Instrument Cluster
2. Anti-Lock Brake System Module
3. Restraints Control Module
4. Occupant Classification Module
5. Parking Aid Module
6. Power Steering Control Module
7. Powertrain Control Module
8. Accessory Protocol Interface Module (SYNC)

The MS CAN bus contains the following components

9. Instrument Cluster
10. Audio Control Module
11. HVAC Module
12. Front Controls Interface Module
13. Front Display Module
14. Smart Junction Box
15. Accessory Protocol Interface Module (SYNC)

Notice that the Instrument Cluster and Accessory Protocol Interface Module bridge the two networks.

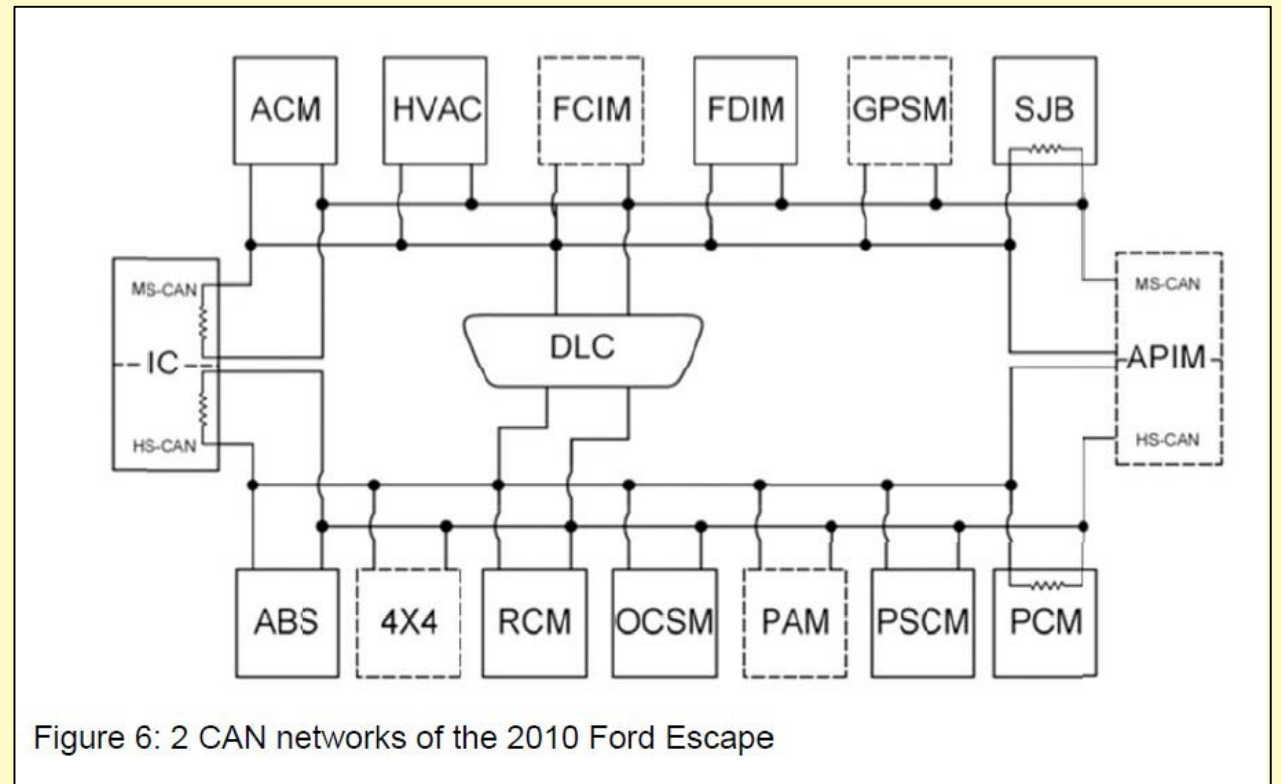


Figure 6: 2 CAN networks of the 2010 Ford Escape

# Notes from: *Adventures in Automotive Networks and Control Units*

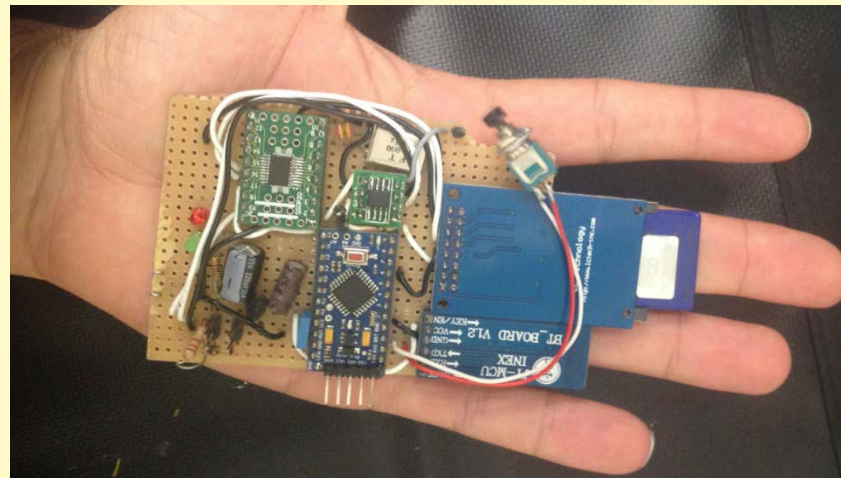
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## Attacks via the CAN bus - Normal packets

- Some examples that can affect the functioning of the automobile by sending normal CAN packets
  - If an attacker could get code running on an ECU (via an attack over Bluetooth, telematics, tire sensor, physical access), they would be able to send these packets and thus to make the car perform these actions.
  - Speedometer – Ford - setting the speed and RPM displayed to the driver
  - Odometer – Ford – can make the Odometer go up
  - Limited steering – Ford - possible to overload the CAN network causing a DoS on the CAN bus
    - No CAN messages can be delivered
    - Different ECUs act differently
    - The PSCM ECU completely shuts down → causes it to no longer provide assistance when steering
    - The wheel becomes difficult to move and will not move more than around 45% no matter how hard you try
    - This means a vehicle attacked in this way can no longer make sharp turns but can only make gradual turns
  - Steering – Ford - the ability to steer the wheel to any position

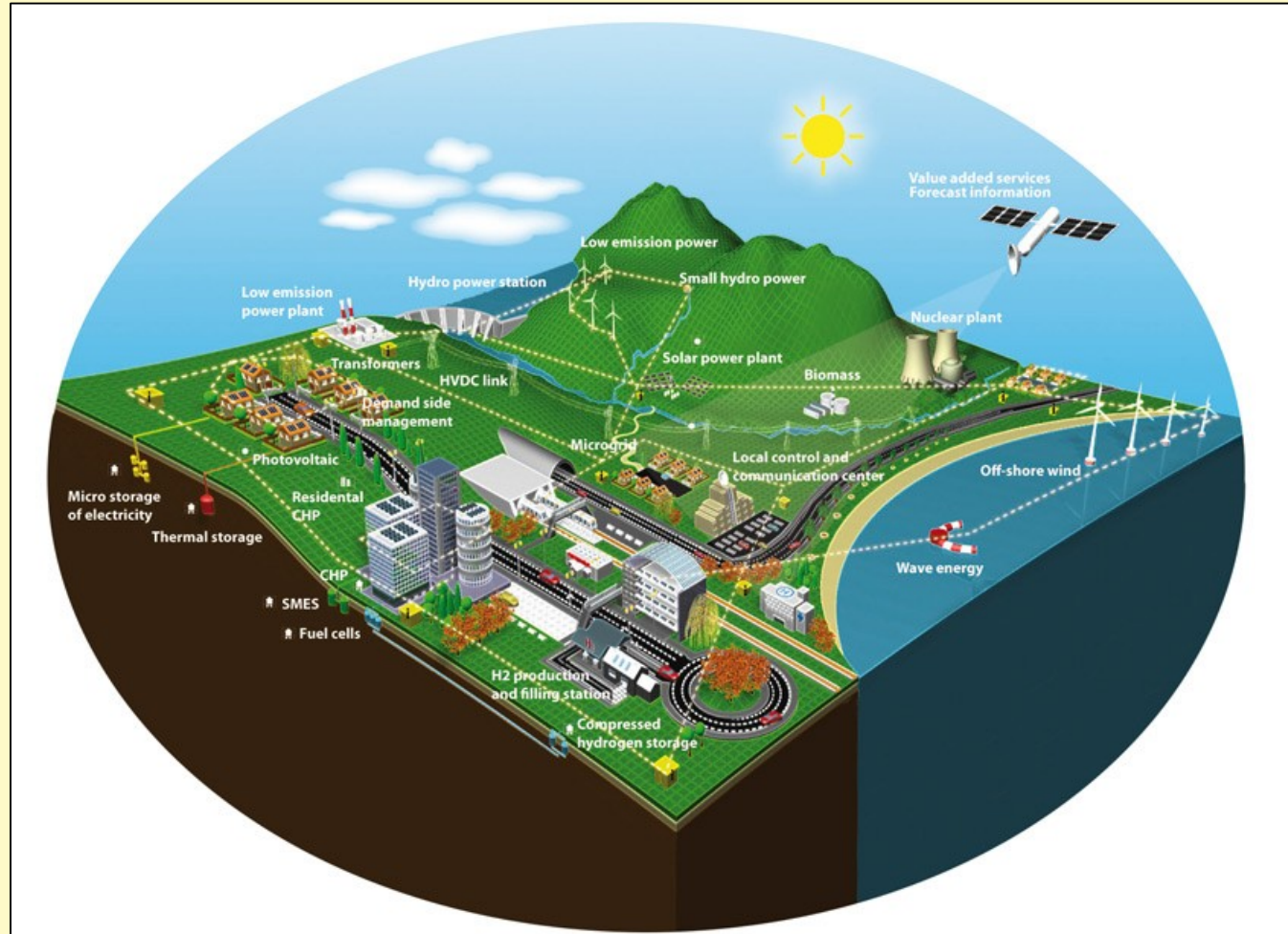
# The Automobile – Hacked ... Again

- Auto makers have long downplayed the threat of hacker attacks on their cars and trucks, arguing that their vehicles' increasingly-networked systems are protected from rogue wireless intrusion
- Two researchers plan to show that a few minutes alone with a car and a tiny, cheap device can give digital saboteurs all the wireless control they need.
- Spanish security researchers Javier Vazquez-Vidal and Alberto Garcia Illera plan to present a small gadget they built for less than \$20 that can be physically connected to a car's internal network to inject malicious commands affecting everything from its windows and headlights to its steering and brakes
- Their tool, which is about three-quarters the size of an iPhone, attaches via four wires to the Controller Area Network or CAN bus of a vehicle, drawing power from the car's electrical system and waiting to relay wireless commands sent remotely from an attacker's computer. They call their creation the CAN Hacking Tool, or CHT.



# What does Cyberspace look like?

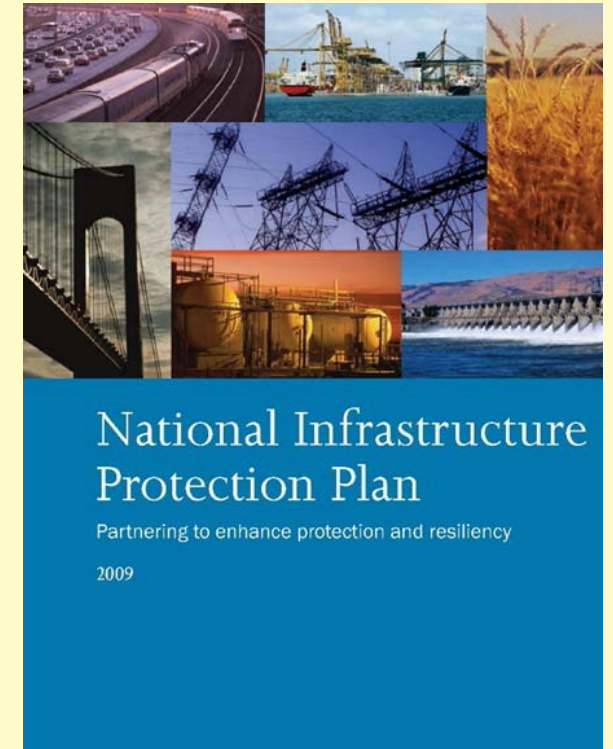
## Critical Infrastructure



# Critical Infrastructure & Key Resources (CIKR)

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- **Critical Infrastructure:** Systems and assets, whether physical or virtual, so vital to the United States that the incapacitation or destruction of such systems and assets would have a debilitating impact on national security, national economic security, public health or safety, or any combination of those matters
- **Key Resources:** Publicly or privately controlled resources essential to the minimal operations of the economy or government
- **Why is CIKR Protection Important?**
  - Essential to the Nation's security, public health and safety, economic vitality, and way of life



Homeland  
Security



# 18 sectors

## CRITICAL INFRASTRUCTURE SECTORS



Agriculture  
and Food



Banking  
and Finance



Chemical



Commercial  
Facilities



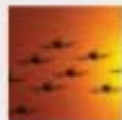
Communications



Critical  
Manufacturing



Dams



Defense Industrial  
Base



Emergency Services



Energy



Government  
Facilities



Healthcare and  
Public Health



Information  
Technology



National  
Monuments  
and Icons



Nuclear Reactors,  
Materials  
and Waste



Postal and Shipping



Transportation  
Systems



Water

Source: [http://www.dhs.gov/files/programs/gc\\_1189168948944.shtm](http://www.dhs.gov/files/programs/gc_1189168948944.shtm)

# TTU Workshop

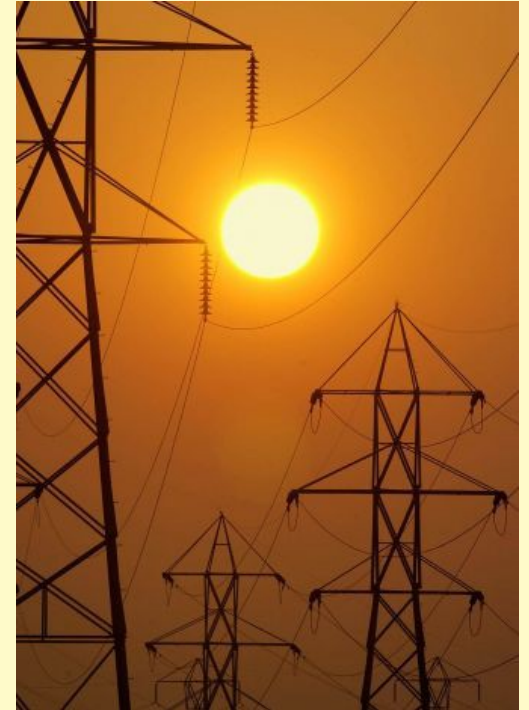
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- Attended Workshop at TTU Feb 2016
  - External Advisory Committee for CyberSecurity
    - CyberSecurity of Industrial Control Systems program
  - Membership
    - AFRL
    - AEP
    - Shell Oil
    - IBM
    - Raytheon
    - Oncor Electric Delivery
    - National Instruments
  - Topics
    - CybSec of Smart Grid Systems
    - Resiliency of Critical Infrastructures
    - Countering Threats to Smart Cities

# Power Grid Protocols

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- The power industry also has its share of protocols that are utilized for interoperability
- In North America, Distributed Network Protocol 3 (DNP3) is used frequently in process automation for electric utilities
- DNP3 is built on top of the Internet protocols and supports two-way communications for control centers, remote terminal units (RTUs) and intelligent electronic devices (IEDs)
- DNP3 defines the security model for proper message authentication and encryption between end points
- In Europe, the IEC 61850 protocol has been adopted and has similar DNP3 functionality and characteristics
- Although there are other protocols for electric systems, these two technologies are well-established and will be considered when integrating existing infrastructure with tomorrow's grid.



# What is the Smart Grid

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A smart grid is an electrical grid which includes a variety of operational and energy measures including smart meters, smart appliances, renewable energy resources, and energy efficiency resources. Electronic power conditioning and control of the production and distribution of electricity are important aspects of the smart grid. [https://en.wikipedia.org/wiki/Smart\\_grid](https://en.wikipedia.org/wiki/Smart_grid)



# Security Challenges for the Smart Grid

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## Security of Power Grid and Smart Grid

- Traditionally security to power system means – to withstand unexpected disturbances
  - Such as short circuit
  - Loss of a power system component such as Transmission line
- In today's world security focus has expanded to include
  - disturbances due to overloading or unexpected causes
    - Physical attacks or
    - Cyber attacks



# Security Challenges for the Smart Grid

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## The Biggest Myth!!

- “We are secure because we are isolated from the Internet and other networks”.
- After Stuxnet in Iran ....
  - Myth gone haywire....
- Its only a matter of time!!
  - Social Engineering => Sneakernet
- Cyber breach will not effect us as we are not controlling from remote.

# Look Familiar ???

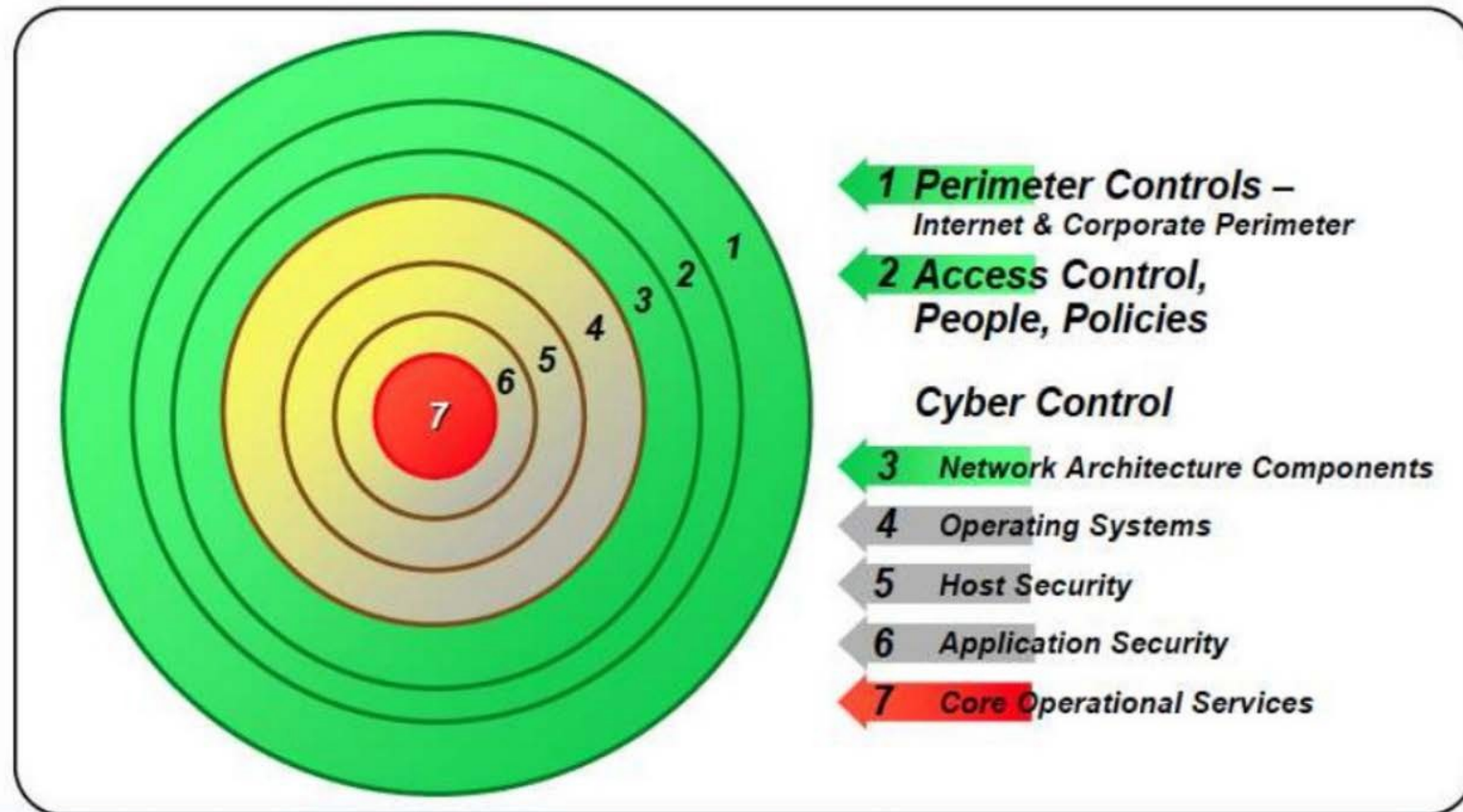
## Understanding Security Exposures...

### *Three-Step Process*



# Defense in Depth

## Defense in-Depth Security



# Security Challenges for the Smart Grid

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## Cyber-Physical Approach to Smart Grid Security

- Physical systems operated are monitored, coordinated, controlled by a computing and communication core.
- Computing and communication capabilities will soon be embedded in all types of objects and structures in the physical environment.
- Smart grid will have more and more such embedded objects.
- Protecting critical infrastructure is vital to the health of an economy;
- one such infrastructure, the electric power transmission grid, forms one of the largest complex nterconnected networks ever built.



# Security Challenges for the Smart Grid

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## Security of Cyber-Physical System

- Tight coupling between ICT and physical system introduces new security concerns and requires a rethinking to common security approach.
- The smart grid will reach every house and building, giving potential attackers easy access to some of the grid components.
- A coordinated assessment of cyber and physical risks keeping the whole grid security goals in mind is needed.
- Bringing together cyber security and system theory is needed to address the security requirements.
  - Cyber attacks can cause disruptions that transcend the cyber realm and affect the physical world –Stuxnet.
  - Physical attacks can affect the cyber system - integrity of a meter can be compromised by using a shunt to bypass it